REPORT LAB 10:

DETERMINING THE C_p/C_v RATIO OF GAS

Class: CC10 / Group: 1		Lecturer's comment
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I. Aims/Purposes

- Investigating the mechanisms that alter the thermodynamic equilibrium of air in a vase and calculating the molecular heat capacity ratio of the gas.

II. Apparatus, Methods, and Procedure

a) *Apparatus*

- Large glass bottle with capacity 10L
- U-shape water manometer with millimeter-scale with accuracy 1mm
- Technical Brass Gas Ball Valve K1, K2
- Rubber ball pump
- Support stand

b) *Methods*

- We will use the Clement and Desormes method to determine the ratio of specific heats of air at constant pressure and constant volume. We will perform an adiabatic expansion on a set amount of air and measure its volume, pressure thereby determining the C_p/C_v ratio

c) *Procedure*

Step 1:

- Learn the opening and closing positions of valves K1 and K2 by observation.

Step 2:

- Close the valve K_2 and open the valve K_1.
- Pump air into flask A until the difference in the water column in the two branches of the manometer M reaches about 250÷300 mmH_2O.
- Close valve K_1.
- Wait 4÷5 minutes for the temperature and pressure of the gas in the flask to reach a stable equilibrium.
- Measure many times with the same initial pressure p_1.

- Slowly open valve K_2 to reduce the amount of air in flask A so that the difference in the water column in the two manometer branches M reaches the preselected value $H = L_1 + L_2$ (in the range of 250÷300 mmH2O).
- Read the values of L_1, L_2 on the millimeter ruler of the manometer holder M to determine the value of H and record it in Table 1.

Step 3:
- Quickly open valve K_2 to let the air in flask A escape.
- When the air pressure in flask A equals the atmospheric pressure H_0 outside, we quickly close valve K_2.
- For accurate measurement results, we need to observe and quickly close valve K_2 as soon as the water level in the two branches of the manometer M has just reached the same level, combined with the sound of air escaping from vase A has just stopped.
- Wait 4÷5 minutes for the temperature of the gas in flask A to equilibrate to room temperature.
- Then, the water head difference in the two manometer branches M reaches a stable value $h = L_1' + L_2'$. Read the values of L_1', L_2' on the millimeter ruler of the manometer holder M to determine the value of h and record it in Table 1.

Step 4:
- Repeat step 2 and step 3 five times, corresponding to (or approximately) the selected value of H.
- Record the corresponding measurement results for h in each measurement in Table 1.

III. Equations

a) *Calculating the theoretical value of the γ ratio of air according to the formula:*

$$\gamma_{theory} = \frac{i+2}{i}$$

b) *Comparing the measured value $\gamma_{measured}$ measured with the theoretical value γ_{theory} by calculating the relative deviations:*

$$\delta = \frac{\gamma_{theory} - \gamma_{measured}}{\gamma_{theory}}$$

IV. Experimental data

Table 1

Accuracy of millimeter ruler on the water column manometer M: 1 (mm)				
	H (mmH$_2$O)	h (mmH$_2$O)	$\gamma = \frac{H}{H-h}$	$\Delta\gamma$
1	255	74	1.409	0.003
2	255	73	1.401	0.005
3	256	74	1.407	0.001
4	250	71	1.397	0.009
5	252	73	1.408	0.002
6	268	78	1.411	0.005
7	280	82	1.414	0.008
8	273	79	1.407	0.001
9	296	86	1.410	0.004
10	272	78	1.402	0.004
11	251	72	1.402	0.004
Average			1.406	0.004

V. Calculation work

a) Calculating the average value:

$$\bar{\gamma} = \frac{1.409+1.401+1.407+1.397+1.408+1.411+1.414+1.407+1.410+1.402+1.402}{11} = 1.406$$

b) Calculating the average absolute error:

$$\overline{\Delta\gamma} = \frac{0.003+0.005+0.001+0.009+0.002+0.005+0.008+0.001+0.004+0.004+0.004}{11} = 0.004$$

c) Results:

$$\gamma = \bar{\gamma} \pm \Delta\gamma = 1.406 \pm 0.004$$

d) Calculating the theoretical value of the γ ratio of air according to the formula:

$$\gamma_{theory} = \frac{i+2}{i} = \frac{6+2}{6} = 1.333$$

e) Comparing the measured value $\gamma_{measured}$ measured with the theoretical value γ_{theory} by calculating the relative deviations:

$$\delta = \frac{\gamma_{theory} - \gamma_{measured}}{\gamma_{theory}} = \frac{|1.333 - 1.406|}{1.333} = 0.055$$

VI. Question

1. Define and write the expressions for specific heat and molecular heat capacity. Does the heat capacity of a gas depend on the conditions of the heating process?

The state of a gas mass is characterized by the following geometries: pressure p, volume V, temperature T.

For 1 mole of gas, these quantities are related by the equation:

$$pV=RT$$

Here R=8.31 J/mol.K is the gas constant.

When heat is transferred to the air in the presence of mass m, the air mass will heat up and its temperature donates an additional amount of di. The specific heat capacity of a substance when it is a quantity measured by the amount of heat that needs to be transferred to 1 kilogram of the substance when its temperature is increased by 1 degree. Therefore:

$$C_s = \frac{\delta.Q}{mdT}$$

For the molecular heat capacity of the gas, the formula is:

$$C_M = \mu.C_s = \frac{\delta.Q}{dT}$$

The unit of measurement for C_s is J/kg.K, of C_M is J/mol.K and of μ (molar mass) is kg/mol. The heat capacity of a gas depends on the characteristics of the heating process (different pressure or different volume).

2. Distinguish between isothermal molecular heat capacity C_v and isobaric molecular heat capacity C_p. Find a relation between them to show that $C_p > C_v$.

C_v is the heat capacity during the transformation where the volume of the system remains constant and is expressed in cells $\delta.Q_v$ divided by n.dT.

C_p is the heat capacity calculated during the transformation process where the pressure of the system remains constant and is expressed in cells $\delta.Q_p$ divided by n.dT.

The related formulas have been established in the measurement principle section (Equation 8).

3. In fact, under what circumstances can gas compression or expansion processes be approximated as isothermal or adiabatic? After compressing or expanding the gas contained in vase *A*, why does it take a certain amount of time (*about 4÷5 minutes*) for the water column difference on the two branches of the manometer M to reach a stable value?

- When the process of compression or expansion of the gas happens very quickly, it does not have time to exchange heat with the outside (Q=0), so it can be approximated as the process of compression or adiabatic expansion. When the process of compression or expansion of a gas takes place for a long time (enough for thermal equilibrium), it is considered isothermal compression or expansion (because the ambient temperature is assumed to be constant and the gas will be in equilibrium with the temperature of the gas). environment).

- Allow a period of 4-5 minutes for the temperature of the gas in flask A to equilibrate to room temperature. Then the water column difference in the two manometer arms M reaches a stable value.

4. Why in this experiment a water column manometer must be used instead of a mercury manometer for measuring the gas pressure in glass flask A?

- In this experiment, we must use a water column manometer but not a mercury manometer to measure the gas pressure in glass vessel A because:

- We have experimental conditions as: h and H<<H_0 . We have H_0 of Hg = 760 mm, H_0 of water is 10.3m. This demonstrates that measuring pressure with water is more sensitive than mercury. Therefore, if we use mercury when we measure the values of h, H, it will be difficult to observe the change and measure H, h.

- Experimental safety measurement: Mercury is a very toxic substance, our measuring tube is a U-shaped tube with two open ends, if we let it fall out, it is very dangerous!

5. To ensure accurate measurement results, why do we have to close valve K_2 as soon as the water column in the two manometer branches M has just reached the same level?

- To ensure accurate measurement results, first, we must close valve K2 quickly to make sure the process is adiabatic (no transference of heat or mass to the surrounding. If we act slowly, there will be transference which will skew our results). Secondly, the reason that we need to close K2 as soon as the water column in the manometer equalize is because this is exactly when the internal pressure of the gas equals the eternal pressure (the rustling sound ceases at this exact moment). At this moment, p2 = H0 and we must close the valve here to ensure accurate readings.

6. Prove the formula for calculating the relative error of the ratio of the molecular heat capacity of a gas $\gamma = \frac{H}{H-h}$ of the form:

$$\frac{\Delta\gamma}{\gamma} = \frac{\boldsymbol{H.\Delta h + h.\Delta H}}{\boldsymbol{H.(H-h)}}$$

We have the formula:

$$\gamma = \frac{H}{H-h}$$

Take the natural logarithm on both sides, we get

$$ln(\gamma) = ln(H) - ln(H-h)$$

Then take the differential:

$$\frac{\Delta\gamma}{\gamma} = \frac{\boldsymbol{\Delta H}}{\boldsymbol{H}} - \frac{\boldsymbol{\Delta H - \Delta h}}{\boldsymbol{H-h}} \leftrightarrow \frac{\Delta\gamma}{\gamma} = \frac{\boldsymbol{-h.\Delta h + H.\Delta h}}{\boldsymbol{H.(H-h)}}$$

7. Calculate the theoretical value of the ratio of the molecular heat capacity of dry air molecules (assume only oxygen molecules O_2 and nitrogen N_2) to the number of degrees of freedom i of the gas molecules.

Since O_2 and N_2 are both diatomic gases, i=5.

$$Cv = \frac{i}{2}R = \frac{5}{2}R$$

$$Cp = Cv + R = \frac{5}{2}R + R = \frac{7}{2}R$$

$$\rightarrow \gamma = \frac{Cp}{Cv} = \frac{7}{5} = 1{,}4$$

8. If the air in the flask has high humidity, contains a lot of water vapor, how will the theoretical value of the ratio of the molecular heat capacity of the air change (increase or decrease compared to dry air)? Why? Knowing that the degrees of freedom i depend on the structure of gas molecules: monoatomic molecule (inert gas): i = 3, diatomic molecule (O_2, N_2…): i = 5, polyatomic molecule (H_2O, CO_2...): i = 6.

Use the formula:

$$\gamma = \frac{Cp}{Cv} = \frac{i+2}{i} = 1 + \frac{2}{i}$$

From the above formula, we can see that the theoretical value of the molecular heat capacity ratio of humid air will decrease compared to dry air because the humid air contains H_2O which i_{H2O}=6 so i of the system is not humid air will be larger than i of dry air system: γ decreases.

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